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TCFD Compliance and Decarbonization Practices: Evidence From Taiwan's Financial Institutions

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ABSTRACT

Using Taiwan's unique Task Force on Climate-Related Financial Disclosures (TCFD) compliance dataset on 80 financial institutions compiled based on a content analysis methodology, we find that financial firms' extent of decarbonization practices positively affects TCFD compliance, suggesting that firms with more substantive decarbonization commitments provide more robust climate-related financial disclosures, effectively articulating their efforts and commitments to stakeholders. Moreover, enhanced TCFD compliance positively influences firms' decarbonization practices in terms of Science Based Targets initiative (SBTi) participation, Scope 3 emissions disclosure, and reduced Scope 1 and 2 emissions intensity. We attribute the positive relationship between TCFD compliance and decarbonization practices to heightened stakeholder scrutiny under Taiwan's mandatory disclosure regime. By increasing public awareness of firms' disclosure and decarbonization practices, TCFD compliance is likely to be a genuine indicator of decarbonization commitments. Our findings offer insights for other emerging markets considering the adoption of mandatory climate-related financial disclosure.

1 | Introduction

This study aims to determine whether a higher degree of climate-related financial disclosures signal improves carbon performance, or whether it is merely a symbolic gesture by high-carbon emitters seeking legitimacy from stakeholders. Alternatively, do improved climate-related financial disclosures reflect a genuine commitment to decarbonization, or is it only “cheap talk” devoid of substantive action? The demand for transparent climate-related financial disclosures is increasing owing to the financial materiality of climate risks and the potential opportunities they present to firms. In response to this trend, as well as the 2015 Paris Agreement, the Financial Stability Board (FSB) and Group of 20 established the task force on climate-related financial disclosures (TCFD) in 2015. Firms worldwide have adopted the framework and recommendations developed by

TCFD. Although TCFD disbanded in 2023, the International Sustainability Standards Board (ISSB) of International Financial Reporting Standards (IFRS) still adopted its framework and recommendations in the development of IFRS S2: Climate-related Disclosures. In Taiwan, aligning with TCFD recommendations, the Financial Supervisory Commission (FSC) issued “Guidelines for Domestic Banks' Climate Risk Financial Disclosure” and the “Guidelines on Climate-related Financial Disclosures of Insurance Companies” to guide financial institutions in reporting climate-related financial information. Under the FSC's Green Finance Action Plan 2.0, published in 2022, both banks and insurers have been mandated to disclose climate-related financial information starting in 2023 (FSC 2022). In sum, the increased demand for transparency and evolving regulatory requirements drive the growing importance of climate-related financial disclosure practices.

[Correction added on 18 December 2025, after first online publication: In the author byline, affiliation 1 has been removed from Yung-Ming Shiu, and affiliation 2 has been updated to start with “Department of Risk Management and Insurance” in this version.]

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Previous studies on TCFD compliance have primarily examined the methodologies employed to construct the TCFD compliance index, including content analysis (e.g., Demaria and Rigot 2021; Cosma et al. 2022; Braasch and Velte 2023; Di Marco et al. 2023; Friedrich et al. 2023; Moses et al. 2025) and text mining (Moreno and Caminero 2022; Ding et al. 2023). Among these studies, Cosma et al. (2022), Di Marco et al. (2023), Ding et al. (2023), Friedrich et al. (2023), and Moses et al. (2025) further analyze the determinants of TCFD compliance, often incorporating explanatory variables such as stakeholder pressure, corporate governance, and other firm characteristics. Regarding sustainability disclosures in a broader context, previous studies have documented findings related to the association between voluntary sustainability disclosures and environmental performance (Al-Tuwaijri et al. 2004; Clarkson et al. 2008; Luo and Tang 2014; Luo 2019).

Despite the growing importance of climate-related financial disclosures and relevant studies, in-depth research and discussions on the link between TCFD compliance and decarbonization commitment remain limited. From the signaling theory perspective, firms with genuine sustainability practices are likely to enhance TCFD compliance to differentiate themselves from others. Moreover, from the perspective of legitimacy and institutional theories, firms with better TCFD compliance are likely to demonstrate genuine sustainability practices to gain legitimacy and conform to the norms in their institutional environments. Regarding practical implications, Taiwan's financial institutions are required to provide climate-related financial reports, and the mandatory disclosure regime may further strengthen this positive relationship because of increased public attention and stakeholders' pressures on firms' disclosure and sustainability practices. To the best of our knowledge, only Ding et al. (2023) and Moses et al. (2025) include carbon emissions variables when analyzing the TCFD compliance level, and no previous study explicitly discusses and examines the bi-directional positive relationship between TCFD compliance and decarbonization commitment implied by the above-mentioned theories. Moreover, scarce studies are based on the sample countries with mandatory disclosure regimes. This study aims to fill in these gaps.

Moreover, our study complements studies on the Global Reporting Initiative (GRI) and other sustainability disclosures in a broader context (e.g., Haque and Ntim 2018; Luo and Tang 2023). Although the TCFD disbanded in 2023 after fulfilling its remit, and the IFRS Foundation published IFRS S2 Climate-related Disclosures to offer guidance on firms' climate-related financial disclosures, the importance of the TCFD remains because the IFRS sustainability disclosures (as well as the Task Force on Nature-Related Financial Disclosures [TNFD]) still utilize the four core elements of reporting: governance, strategy, risk management, and metrics and targets. In addition, GRI reporting primarily aims to disclose the impacts of firms' activities on sustainability themes, whereas TCFD focuses on the financial impacts of climate-related risks and opportunities on firms. Thus, the target audience for GRI reporting is stakeholders (mainly non-investors), whereas the target audience for TCFD reporting is investors. Furthermore, based on the Consolidated Set of the GRI Standards, climate change is an important sustainability theme in GRI, and the transition plan,

scenario analysis, and carbon emission disclosures are highly relevant to TCFD recommended disclosures (Global Reporting Initiative 2025).

Accordingly, this study's purpose is twofold. First, we analyze how decarbonization practices variables—such as participation in the science-based targets initiative (SBTi), disclosure of Scope 3 emissions, and Scope 1 and 2 emissions intensity—affect TCFD compliance. Second, we investigate whether a high TCFD compliance level corresponds to increased decarbonization commitment. Specifically, we examine the impact of TCFD compliance on the aforementioned decarbonization practices and commitment variables. We utilize the proprietary data on financial institutions' TCFD compliance scores from the first and second annual climate-related financial disclosure assessment of Taiwan's financial institutions conducted by the Center for Business Sustainability at National Chengchi University (NCCU) in 2023 and 2024.

The study uses Taiwan's sample for two main reasons. First, previous studies on TCFD compliance concentrate on developed countries such as the UK, US, Canada, and Australia (Ding et al. 2023); the UK, Australia, and New Zealand (Moses et al. 2025); Spain (Moreno and Caminero 2022); and the European financial sector (Di Marco et al. 2023; Friedrich et al. 2023). By contrast, relevant studies on TCFD compliance in the context of emerging markets remain scant. According to jurisdictional snapshots issued by IFRS, the Ministry of Finance (MoF) in China is establishing its climate-related disclosure standard by 2027 (IFRS Sustainability 2025a). Similarly, Korean authorities plan to adopt sustainability disclosure standards after 2026, and climate-related financial disclosure will be mandatory in the near future (IFRS Sustainability 2025b). Compared to developed countries, emerging markets seem to lag in adopting climate-related financial disclosure requirements. As Taiwan is considered an emerging market, our study can offer insights into industry practices and future research in the context of other countries considered emerging markets. Second, financial institutions' TCFD compliance scores are determined by annual climate-related financial disclosure assessments, which surpass the control of those financial institutions. Such a feature helps mitigate endogeneity concerns that may arise from the interplay between TCFD compliance, decarbonization commitment, and other firm characteristic variables in the empirical analyses.

We analyzed a sample of 80 financial institutions in Taiwan and documented the findings. First, we analyzed the determinants of TCFD compliance level and found that sound environmental practices in terms of the participation in SBTi, Scope 3 emission disclosure, and reduced Scope 1 and 2 emissions intensity, lead to better TCFD compliance levels. In extension, our results suggest that higher-quality climate-related financial disclosure may serve as a channel for firms to increase their awareness of climate-related issues and to communicate their efforts and actions to stakeholders. Second, we found that greater TCFD compliance results in increased likelihood of SBTi participation, Scope 3 disclosure, and reduced Scope 1 and 2 emissions intensity, suggesting that enhanced climate-related financial disclosures signal firms' substantive intention to decarbonize. Collectively, this study extends previous research on the relationship between voluntary sustainability disclosures and

environmental performance by documenting the positive and bidirectional relationships between TCFD compliance and decarbonization practices.

2 | Hypotheses Development

First, we discuss the effects of decarbonization practices on TCFD compliance levels. From a greenwashing concern perspective, decoupling refers to the gap between the sustainability information conveyed to stakeholders and actual sustainability performance (Zioło et al. 2024; Keresztúri et al. 2025). Combining the greenwashing and legitimacy theory perspectives, climate reporting may serve as a means for firms without sound environmental performance to safeguard their reputations and garner stakeholder support (Ding et al. 2023; Friedrich et al. 2023; Orazalin et al. 2024). Consistent with this argument, Luo (2019) and Ding et al. (2023) suggest that firms' carbon emissions are positively associated with voluntary carbon disclosure and the quality of climate-related financial information disclosure. Collectively, these studies imply that firms with lower decarbonization efforts may exhibit higher TCFD compliance levels to seek legitimacy from stakeholders.

However, when the arguments of information asymmetry and signaling theory are combined, for firms with sound decarbonization practices, enhancing TCFD compliance can serve as a signal to overcome the adverse selection problem associated with information asymmetry. As Benameur et al. (2024) suggest, sustainability reporting is a crucial way for firms to communicate sustainability commitments to stakeholders. From the greenwashing perspective, responsible investors and other external parties may not manage to discern the firms engaging in greenwashing and those with genuine environmental efforts and performance (Valero-Gil et al. 2024; Keresztúri et al. 2025). For firms with sound sustainability practices, enhancing sustainability disclosure to external stakeholders may signal genuine sustainability commitment and performance, helping to mitigate this problem. For instance, Salvi et al. (2025) suggest that climate change disclosure serves as a tool for firms to signal external investors to reduce information asymmetry and the cost of capital.

Consistent with these arguments, Al-Tuwaijri et al. (2004) argue that firms with strong environmental performance tend to proactively disclose such performance to investors and stakeholders, potentially enhancing their valuation owing to reduced legal risks and environmental liabilities. Empirical evidence supports this: Al-Tuwaijri et al. (2004), Clarkson et al. (2008), and Luo and Tang (2014) document a positive correlation between environmental performance and disclosure. Regarding the evidence in the government sector, Nicolò et al. (2024) find that Italian local governments that perform better in Sustainable Development Goal (SDG) 13 are more inclined to disclose climate-related information to convey their genuine efforts to enhance climate resiliency and decarbonization performance.

The increased stakeholder scrutiny on firms under the mandatory disclosure regime also supports the positive effect of

decarbonization practices on TCFD compliance. For example, Fiechter et al. (2022) and Aboud et al. (2024) demonstrate that the enactment of EU Directive 2014/95, which mandates non-financial reporting, significantly improves firms' CSR initiatives and reduces ESG decoupling. Specifically, Fiechter et al. (2022) suggest that the passage of the Directive increases public attention and stakeholders' pressures on firms' CSR disclosure and practices, and once CSR transparency increases, stakeholders further pressure firms' CSR practices. Moreover, Aboud et al. (2024) suggest that the passage of the Directive changes firms' institutional environment and introduces more coercive and normative pressures on firms because of new laws and rules. These studies imply that firms face greater stakeholder scrutiny under the mandatory disclosure regime. In addition, Forcadell et al. (2023) further suggest that increased stakeholder scrutiny heightens the credibility of CSR signals. Taken together, owing to the increased stakeholder scrutiny under the mandatory disclosure regime, firms' CSR/ESG disclosures are more likely to be consistent with their real practices.

In 2023, Taiwan's FSC issued "Guidelines for Domestic Banks' Climate Risk Financial Disclosure" and the "Guidelines on Climate-related Financial Disclosures of Insurance Companies" to mandate climate-related financial disclosure for financial institutions. Owing to the regulatory pressure from these mandatory disclosure requirements, climate-related financial disclosures are increasingly aligned with decarbonization practices, suggesting a positive relationship between decarbonization efforts and TCFD guidelines. Thus, we propose the following research hypothesis:

Hypothesis 1. *The extent of decarbonization practices engagement positively affects the level of TCFD compliance.*

From an alternative analytical perspective, this study investigates whether higher levels of compliance with TCFD recommendations enhance firms' commitment to decarbonization. For instance, Binger et al. (2022, 2024) argue that climate reporting may serve as a form of "cheap talk" without any substantive goals or genuine commitments to decarbonization. Moreover, Ratti et al. (2025) analyze the effects of various climate disclosures on direct and indirect emissions, which refer to Scopes 1 and 2 emissions, respectively. They find that, in the short term, these disclosures lead to a reduction in Scope 2 emission but no significant reduction in direct emissions. In the medium term (i.e., 2 years or more), only firms with clear carbon neutrality goals exhibit significant reduction in direct emissions, while other forms of climate disclosure have no significant impact on direct emissions. Their findings raise concerns about greenwashing, as they suggest a decoupling between climate-related disclosures and actual environmental practices. Taken together, if a high level of compliance with TCFD recommendations is merely symbolic "cheap talk," we would expect that such compliance does not improve decarbonization practices.

Nevertheless, combining legitimacy and institutional theories, firms' actions should conform to the external rules, norms, laws, and best practices in their institutional environments to gain legitimacy and stakeholders' support (e.g., Eliwa

et al. 2021). Making sustainability reporting mandatory can exert institutional pressures on firms' ESG practices, thereby reducing the extent of ESG decoupling to gain legitimacy (Fiechter et al. 2022; Aboud et al. 2024). For instance, Fiechter et al. (2022) suggest that mandating CSR disclosure improves both CSR disclosures and practices. In their study, the factors contributing to the positive effect of the passage of the CSR Directive in 2014 on CSR activities include the pressures from stakeholders and industry peers, firms' internal education and preparations for the mandatory reporting, and increased public attention on firms' CSR activities. Furthermore, as Aboud et al. (2024) suggest, mandating ESG reporting changes the business environment and exerts coercive and normative pressures on firms to adopt best ESG practices to conform to the norms and requirements within the institutional environments in which they operate. In response to these pressures, firms are likely to improve their CSR/ESG practices, and thus reduce the extent of greenwashing, to gain legitimacy in their institutional environments.

We review other relevant studies supporting the positive effect of TCFD compliance on decarbonization practices as follows. First, Braasch and Velte (2023) contend that high-quality disclosure is less likely to be symbolic, and firms' carbon performance is likely to improve as the quality of climate reporting increases, primarily owing to stakeholder feedback. Second, previous studies indicate that self-regulatory global sustainability institutions, such as the Global Reporting Initiative (GRI), function as either complements to or substitutes for laws and regulations on sustainability (Haque and Ntim 2018; Luo and Tang 2023). Specifically, the adoption of sustainability frameworks developed by these self-regulatory global organizations can demonstrate firms' genuine commitment to decarbonization. Findings from these studies suggest that firms implementing the GRI framework in sustainability and ESG reporting tend to engage in more substantial carbon mitigation efforts.

Third, the recommendations provided by the TCFD offer a structured set of guidelines for firms to disclose climate-related financial information in a transparent, comprehensive, and comparable manner. For instance, Privato et al. (2024) argue that the TCFD framework serves as an optimal tool for firms to formulate well-defined climate transition plans (CTPs), which include components such as risk governance, financial strategies for managing climate risks and opportunities, concrete actions taken, and carbon accounting procedures. Moreover, Morrison et al. (2024) find that firms whose carbon emission targets are set in accordance with TCFD recommendations experience the highest emission reductions.

Collectively, greater adherence to the TCFD recommendations may signal a firm's earnest commitment to decarbonization, which, in turn, can result in improved decarbonization performance. As financial institutions are required to provide climate-related financial reports in Taiwan, we expect that the degree of TCFD compliance is likely to indicate genuine efforts and performance of decarbonization practices. Taken together, our hypothesis is stated as follows:

Hypothesis 2. *The level of TCFD compliance positively affects the extent of decarbonization practices engagement.*

3 | Data and Methodology

3.1 | Data Sources

We obtained the data on the compliance of financial institutions (including banks and insurers) with the TCFD guidelines from 2023 and 2024 assessments conducted by the Center for Business Sustainability at NCCU. Moreover, we gathered the financial data of these banks and insurers from publicly available statistics provided by the Banking Bureau of the FSC and the Taiwan Insurance Institute, respectively. Regarding the information on financial institutions' participation in the SBTi, disclosure of Scope 3 emissions, and their Scope 1 and 2 emissions, we derived the data from their respective TCFD reports. The final sample comprised 80 financial institutions, including 38 domestic banks,¹ 21 life insurers, and 21 non-life insurers,² and a total of 160 firm/year observations.

3.2 | Methodology

3.2.1 | Construction of TCFD Compliance Index

Following prior studies on firms' compliance with the TCFD recommendations, the coding taskforce of the Center for Business Sustainability employed content analysis to evaluate the extent of financial institutions' TCFD compliance (e.g., Demaria and Rigo 2021; Braasch and Velte 2023; Di Marco et al. 2023). Four core elements of the TCFD recommendations—governance, strategy, risk management, and metrics and targets—were identified as key criteria for assessing compliance. In the content analysis, each question was assigned a score ranging from 0 to 3, reflecting the level of compliance. Specifically, scores of 0, 1, 2, and 3 corresponded to different degrees of compliance, with a score of 3 indicating full compliance with the requirements outlined in the question.

In Taiwan, financial institutions compiling their TCFD reports may choose to follow the TCFD recommendations and supplemental guidance for banks, insurers, and asset owners (Task Force on Climate-Related Financial Disclosures 2017, 2021).³ Alternatively, they may adhere to the "Guidelines for Domestic Banks' Climate Risk Financial Disclosure" or "Guidelines for Financial Disclosure of Climate-Related Risks of Insurance Enterprises" issued by the FSC. Consequently, the coding taskforce of the Center for Business Sustainability identifies key questions based on these recommendations and guidelines and develops two fundamental question sets used to assess financial institutions' compliance with TCFD standards. Specifically, the coding taskforce established fundamental question sets containing 39 and 47 questions for banks and insurers, respectively, based on the TCFD recommendations and supplemental guidance. Similarly, for the FSC-issued guidelines, the fundamental question sets included 21 questions for banks and 20 for insurers, respectively.

In addition to the two fundamental question sets mentioned above, the coding task force identified additional bonus questions aimed at promoting best practices in TCFD reporting. Specifically, based on the "TCFD Good Practice Handbook" issued by the Climate Disclosure Standards Board in 2021

(Climate Disclosure Standards Board 2021), the coding task force developed bonus questions for three core elements: strategy, risk management, and metrics and targets.⁴ Similarly, the coding task force assigned a score on a scale of 0 to 3 to each bonus question in the set. If a financial institution's report fully adhered to the criteria outlined in the question, a maximum score of 3 was awarded. These bonus questions were uniformly applied across all financial institutions, regardless of industry sector or the specific recommendations and guidelines they followed when preparing their TCFD reports.

The compliance index was constructed as follows:

$$\text{Normalized Score on each core element} = \left[\frac{\text{The actual score}}{\text{The maximum score}} \right] \times 100 \quad (1)$$

$$\begin{aligned} & \text{TCFD compliance Index} \\ &= \left[\frac{\text{the summation of the scores on four core element from fundamental question set}}{\div 4} \right] \times 90\% \\ &+ \left[\frac{\text{the summation of the scores on three core element from bonus question set}}{\div 3} \right] \times 10\% \\ &+ \text{additional Bonus} - \text{Penalty} \quad (2) \end{aligned}$$

Following Demaria and Rigot (2021) and Braasch and Velte (2023), for each core element, the coding task force calculated the normalized score received by each financial institution by dividing the actual points received by the maximum possible points for this core element, and then multiplying the result by 100, as shown in Equation (1). This step equalizes the weights of all core elements in the calculation of the TCFD compliance index, despite the varying number of questions included in different core elements, and ensures that the scores range from 0 to 100 for each core element. For example, if a core element includes 10 questions and a financial institution receives 21 points for that core element, the maximum possible score would be 30 points (10 questions multiplied by 3 points per question). Therefore, the normalized score for this core element was 70 points (21 divided by 30 points, then multiplied by 100).

In Equation (2), the normalized scores for all core elements are summed, and then a simple average is calculated for both the fundamental and bonus question sets. As the coding task force excluded the governance core element from the scope of the bonus question set, the sum of the scores from the bonus question set is divided by three rather than four. Regarding the weights assigned to the fundamental and bonus question sets, financial institutions' scores on the fundamental set represent their overall compliance. By contrast, the scores on the bonus question set represent additional financial institutions' disclosures—beyond the basic requirements in TCFD recommendations and FSC-issued guidelines. Thus, a significantly higher weight is assigned to the scores on the fundamental question set. To compute the TCFD compliance index, weights of 90% and 10% are assigned to the simple average scores on the fundamental and bonus question sets, respectively.

Furthermore, if financial institutions' TCFD reports receive accreditation from an authoritative third party—such as the British Standards Institution (BSI) or a professional accounting firm—an additional bonus of 1 point is awarded to the financial institution. Financial institutions may also earn additional bonuses through disclosures related to the Taskforce on Nature-Related Financial Disclosures (TNFD) (2023) recommendations. Based on the contents of TCFD reports, if financial institutions disclose additional information relevant to TNFD recommendations, they may earn additional bonus point ranges from 0.2 to 1 point based on the extent of disclosure comprehensiveness. Conversely, if material deficiencies are identified in TCFD reports, such as insignificant inconsistencies with annual ESG or sustainability reports issued by financial institutions, a penalty of 1 point is deducted from the overall TCFD compliance index score.

To reduce subjectivity and enhance consistency in the assessment results of the coders in the coding task force, coding procedures described by Bouten et al. (2011), Demaria and Rigot (2021), and Moses et al. (2025) were implemented before conducting the formal assessment. First, the coding task force conducted a trial assessment on several randomly selected TCFD reports. Each coder independently evaluated the degree of compliance in the selected reports, assigned points for each question, and extracted relevant excerpts and graphs from the TCFD reports for documentation purposes. Next, a peer-checking process was initiated, and for each question, the trial assessment results from the coders were compared upon completing the trial assessment. If there was a discrepancy of two or more points between any two coders for a given question, the coding task force initiated an in-depth discussion to resolve the differences and reach consensus on the approach of reviewing and assessing TCFD reports. During this discussion, the coders referred to the original excerpts and graphs extracted from the TCFD report and explained the rationale for assigning the respective scores. Third, if the discussion revealed inconsistencies or ambiguities regarding the interpretation of the question descriptions, those descriptions were revised to ensure that all coders agreed on their meaning. After completing these trial assessment procedures, the coding task force conducted the formal assessment to evaluate the TCFD compliance levels of financial institutions.

In addition to the formal assessment conducted by the coding task force, the Center for Business Sustainability engaged several experts from relevant fields—such as accounting and finance, risk management and insurance, environmental change, and sustainable development—to evaluate financial institutions' TCFD compliance. Unlike the TCFD compliance index derived in Equation (2), all experts were instructed to assign a score between 60 and 100, reflecting their perception of the institutions' compliance level. External experts were hired to complement the assessments by the coding taskforce. Other than the above-mentioned detailed approach of assigning a score to each question by the taskforce, the experts reviewed the TCFD reports in a more holistic approach, and the scores are determined based on their holistic assessments of the contents of the TCFD reports. The formal assessments by the coding task force and external experts were performed independently, resulting in two distinct sets of evaluation outcomes, which are presented in the subsequent sections.

3.2.2 | Regression Models

We defined the variables representing decarbonization practices in our analyses as follows. First, consistent with Romito et al. (2024) and Ratti et al. (2025), we use the SBTi participation indicator, which is assigned a value of 1 if the firm participated in SBTi and the science-based emission reduction target has been independently verified, and 0 otherwise. As previous studies suggest, in the SBTi participation context, the decarbonization target setting requires third-party validation, which increases the credibility of decarbonization targets and can serve as a signal of substantive commitment to stakeholders (Bendig et al. 2023; Ben-Amar et al. 2024). Thus, we use the participation in SBTi as a variable representing decarbonization practices.

Second, we include the Scope 3 disclosure indicator, taking a value of 1 if the firm reported Scope 3 emissions in the TCFD report and 0 otherwise. Compared with Scope 1 and 2 emissions, verifying Scope 3 emissions is significantly more challenging, as only a limited number of financial institutions disclose these emissions in their TCFD reports. According to the Climate-related Financial Disclosures Yearbook (Center for Business Sustainability at National Chengchi University 2024), the results of a survey on Taiwan's financial institutions suggest that the top reasons for not reporting Scope 3 emissions include the difficulty in collecting credible data to calculate the emissions from investees, financial instruments, and insureds, and inadequate resources devoted to data collection and calculation of Scope 3 carbon emissions (Center for Business Sustainability at National Chengchi University 2024). Owing to the above-mentioned reasons, we consider the disclosure of Scope 3 emissions as an indicator of sound decarbonization practices. Furthermore, the disclosure of Scope 3 emissions is crucial because it can increase awareness among both financial institutions and their stakeholders and may indicate a stronger commitment to mitigating and managing emissions.

Third, following Ratti et al. (2025), we use Scope 1 and 2 emissions intensity as inverse proxies for decarbonization efforts. Specifically, we define Scope 1 and 2 emissions intensity as the ratio of a firm's Scope 1 and 2 emissions (in tons of CO₂ equivalent) to total revenues (\$NT 1 million dollars). Our dataset has 107 firm/year observations with full disclosures of these emissions.⁵

To assess the impact of firms' participation in SBTi and Scope 3 disclosure on the level of TCFD compliance, we constructed Equation (3) as follows:

$$TCFD\ Compliance_i = \alpha + \beta_1 SBTi_i + \beta_2 Scope3_Disclose_i + Firm\ Characteristics_i + \varepsilon_i \quad (3)$$

Among the 107 firm/year observations with fully disclosures of Scope 1 and 2 emissions, we developed Equation (4) to analyze the impact of Scope 1 and 2 emissions levels on TCFD compliance:

$$TCFD\ Compliance_i = \alpha + \beta_1 SBTi_i + \beta_2 Scope3_Disclose_i + \beta_3 Scope12_Intensity_i + Firm\ Characteristics_i + \varepsilon_i \quad (4)$$

In Equations (3) and (4), *TCFD Compliance* represents the level of financial institutions' compliance with the TCFD recommendations. The analysis employed two proxies—the first proxy was the average assessment score derived from three coders within the coding task force, and the second proxy was the average score from assessment results provided by external experts in relevant fields. Turning to the decarbonization variables, *SBTi* and *Scope3_Disclose* denote SBTi participation and disclosure of Scope 3 emissions, respectively, whereas *Scope12_Intensity* represents Scope 1 and 2 emissions intensity. Subscript *i* denotes firm *i*, and ε denotes the error term. Firm Characteristics refers to the control variables capturing firm-specific attributes. Ordinary least squares (OLS) regression was used to estimate Equations (3) and (4).

As predicted in Hypothesis 1, in the analyses of TCFD Compliance, we expect to find positive coefficients on SBTi participation and Scope 3 disclosure and a negative coefficient on Scope 1 and 2 emission intensity. In addition, we identified the control variables representing firm characteristics in Equations (3) and (4) based on Taiwan's institutional background and prior research on the determinants of TCFD compliance and sustainability reporting. The following variables were selected considering Taiwan's institutional context. First, we classify the financial institutions in our sample into three categories: banks, life insurers, and non-life insurers. To represent these three categories, we introduce two dummy variables: *Life_Ins* and *NonLife_Ins*. The variable *Life_Ins* takes a value of 1 if the institution is a life insurer and 0 otherwise, while *NonLife_Ins* is assigned a value of 1 for non-life insurers and 0 otherwise. Second, as some financial institutions are part of financial holding companies, we include an indicator variable, *Fin_Holding*, which equals 1 if the institution is a member of a financial holding company and 0 otherwise.

Third, we included two indicators representing the methods by which financial institutions disclose climate-related financial information. The first indicator is an independent reporting indicator (*Indep_Report*), which takes the value of 1 if firms issue independent TCFD reports and 0 if they report at the financial holding company level rather than at the entity level. The second indicator is the TCFD report indicator (*TCFD_Report*), which equals 1 if a firm discloses climate-related financial information in a TCFD-specific report and 0 if such information is included as part of its annual sustainability report.

In addition to the control variables mentioned above, we identified the following variables that reflect firms' financial resources: firm size (*FSize*), capital holdings (*CapHold*), and profitability (*Profitability*). Based on prior studies on the determinants of TCFD compliance and sustainability reporting, these factors are closely related to the level of resources allocated to report preparation, which may, in turn, influence compliance and reporting quality (Chiu and Wang 2015; Malafronte et al. 2016; Di Marco et al. 2023). Therefore, we included these variables as control variables in our analysis.

Turning to another perspective, to examine the effects of TCFD compliance on participation in the SBTi, the disclosure of Scope 3 emissions, and the volume of Scope 1 and 2 emissions, we constructed Equations (5–7).

$$SBTi = \alpha + \beta_1 TCFD\ Compliance_i + Firm\ Characteristics_i + \epsilon_i \quad (5)$$

$$Scope3_Disclose_i = \alpha + \beta_1 TCFD\ Compliance_i + Firm\ Characteristics_i + \epsilon_i \quad (6)$$

$$Scope12_Intensity_{ii} = \alpha + \beta_1 TCFD\ Compliance_i + Firm\ Characteristics_i + \epsilon_i \quad (7)$$

All definitions of the subscripts and abbreviations are consistent with those in Equations (3) and (4), except for the control variables representing firm characteristics. In Equations (5–7), we incorporate two dummy variables to represent the financial institutions' industry types: life insurer and non-life insurer indicators (*Life_Ins*, *NonLife_Ins*). Additionally, following studies on the determinants of environmental performance (e.g., Luo 2019; Alam et al. 2022; Luo and Tang 2023), we introduce three control variables: firm size (*FSize*), capital holdings (*CapHold*), and profitability (*Profitability*).⁶

To estimate Equations (5) and (6), we employ a logit regression, in which the dependent variable is dichotomous. For Equation (7), as the disclosure of Scope 1 and 2 emissions may be non-random, which may lead to self-selection bias, we adopt Heckman (1979) two-stage regression. In the first step, we estimate financial institutions' decisions to disclose Scope 1 and 2 emissions by employing Probit regression, and Scope 1 and 2 disclose indicator serves as a dependent variable, which equals 1 if Scope 1 and 2 emissions are disclosed, and 0 otherwise. After the first stage of Probit regression estimation, the inverse mills ratio (IMR) is generated and included as an additional explanatory variable in the second stage of the regression analyses. As hypothesized by Hypothesis 2, we anticipate a positive coefficient for TCFD compliance in Equations (5) and (6) and a negative coefficient in Equation (7).

To address the possible endogeneity concerns, in addition to the above-mentioned OLS regression analyses, we also adopt the following measures in our empirical analyses. First, to mitigate the concern arising from simultaneity, we lag the explanatory variables for 1 year, and this measure reduces the sample size from 160 to 80 firm/year observations. Second, we employ the propensity score matching (PSM) method to construct the matched samples for OLS regression analyses. Within each industry type, we match the observations in treated and control groups on a one-on-one basis without replacement. We then perform the regression analyses on these matched samples, and the results are presented in the Extended Analyses section. Table 1 presents a summary of these variables' definitions.

4 | Empirical Results

4.1 | Descriptive Statistics

We provide descriptive statistics for the variables in Table 2. On average, financial institutions received a score of 45.1451 from the coding task force. Beyond the overall TCFD compliance

score, we also analyzed the average scores of the four core elements: governance, strategy, risk management, and metrics and targets. Among these elements, compliance was the lowest in the metrics and targets category, with an average score of 37.2095. One possible explanation for this low compliance is the complexity involved in verifying and disclosing carbon emissions, particularly Scope 3 emissions. Furthermore, data limitations in terms of both quality and availability may contribute to these challenges. Institutions with constrained resources are likely to encounter greater difficulties, resulting in lower levels of TCFD compliance.

Regarding the average score from external experts, the mean value was 80.4403. However, these results should be interpreted with caution. As noted in the previous section, the coding task force and external experts applied distinct procedures and evaluation criteria to generate the assessment results. Consequently, the average scores from these two groups are not directly comparable. For the decarbonization practice variables, the statistics presented in Table 2 indicate that only 26.25% of financial institutions in the sample participated in SBTi, and only 38.13% disclosed Scope 3 emissions. Regarding the correlations between decarbonization practices and TCFD compliance in Table 3, we observe that both SBTi participation and Scope 3 emissions disclosure exhibit a positive association with TCFD compliance. Moreover, among the 107 firm/year observations with Scope 1 and 2 emissions disclosures, we find a negative relationship between the amount of disclosed Scope 1 and 2 emissions intensity and TCFD compliance. Collectively, the correlation coefficients in Table 3 reveal the positive association between TCFD compliance and decarbonization practices.

4.2 | Regression Analyses

4.2.1 | The Effect of Decarbonization Practices on TCFD Compliance

The results shown in Column (1) of Table 4 indicate that both the SBTi and Scope 3 disclosure indicators exert positive effects on the overall TCFD compliance level determined by the coding taskforce, suggesting that financial institutions participating in SBTi and disclosing Scope 3 emissions demonstrate better compliance with the TCFD recommendations or guidance issued by the FSC. Likewise, in Columns (2) to (5) of Table 4, we find that the coefficients on SBTi and Scope 3 disclosure indicators are positive and significant, with only a few exceptions. Regarding the results on the compliance score from external experts, Column (6) of Table 4 shows that firms with SBTi participation have higher-than-average TCFD compliance. Nevertheless, the coefficient on the Scope 3 disclosure indicator is positive but insignificant.

In Table 5, we further incorporate Scope 1 and 2 emission intensity as additional explanatory variables in the analyses of the effects of decarbonization practices on TCFD compliance among the 107 firm/year observations with Scope 1 and 2 emissions disclosure. Our analysis reveals that the majority of the coefficients of the Scope 1 and 2 emissions intensity variable are negative and significant in Table 5, which suggests that firms with lower

TABLE 1 | Variable definitions.

Variable	Definition
Panel A: TCFD compliance variables	
<i>AvgScore</i>	The average score on overall TCFD compliance level by assessment task force members.
<i>AvgScore_Gov</i>	The average score on governance core element of TCFD by assessment task force members.
<i>AvgScore_Str</i>	The average score on strategy core element of TCFD by assessment task force members.
<i>AvgScore_RM</i>	The average score on risk management core element of TCFD by assessment task force members.
<i>AvgScore_MT</i>	The average score on metrics and targets core element of TCFD by assessment task force members.
<i>ExpAvgScore</i>	The average score on overall TCFD compliance level by experts in the relevant fields.
Panel B: Climate initiative and environmental performance variables	
<i>SBTi</i>	If the firm participates in SBTi, the value equals 1, and 0 otherwise.
<i>Scope 3_Disclose</i>	If the firm discloses Scope 3 emissions, the value equals 1, and 0 otherwise.
<i>Scope12_Intensity</i>	The ratio of firm's Scopes 1 and 2 emissions to total revenues.
Panel C: Firm characteristics variables	
<i>Life_Ins</i>	If a firm is a life insurer, the value equals 1, and 0 otherwise.
<i>NonLife_Ins</i>	If a firm is a non-life insurer, the value equals 1, and 0 otherwise.
<i>Fin_Holding</i>	If a firm is a member of the financial holding company, the value equals 1, and 0 otherwise.
<i>Indep_Report</i>	If the firm discloses climate financial information independently, the value equals 1; if the firm discloses climate financial information as a part of the financial holding company's disclosure, the value equals 0.
<i>TCFD_Report</i>	If the firm discloses climate financial information on the TCFD report, the value equals 1; if the firm discloses climate financial information as a part of the annual sustainability report, the value equals 0.
<i>FSize</i>	The logarithm value of total assets.
<i>CapHold</i>	The ratio of equity to total assets.
<i>Profitability</i>	The ratio of pretax income to total assets.

carbon emissions are associated with better TCFD compliance. To synthesize the results in Tables 4 and 5, Hypothesis 1 is generally supported. By enhancing the level of TCFD compliance, financial institutions with sound decarbonization practices can send a credible signal to stakeholders to gain legitimacy and support.

Concerning the results of other control variables, our analysis indicates that firm size consistently exerts a positive effect on TCFD compliance levels, as shown in Tables 4 and 5. This consistent positive relationship may imply that larger firms possess greater resources to improve the quality of climate-related financial disclosures, thereby achieving higher levels of TCFD compliance.

4.2.2 | The Effect of TCFD Compliance on Decarbonization Practices

We present the results of the effects of TCFD compliance on decarbonization practice variables in Tables 6–8. Column (1) of Table 6 indicates that overall TCFD compliance positively influences financial institutions' likelihood of participating in SBTi. Furthermore, TCFD compliance analysis across the four core elements reveals (Columns (2) and (5) of Table 6) that higher average scores in the governance, strategy, risk management, metrics and target elements significantly increase the probability of SBTi participation. Additionally, we observe a positive correlation between external experts' average assigned scores (*ExpAvgScore*) and the likelihood of SBTi participation, as shown in Column (6) of Table 6. Collectively, these findings suggest that improved TCFD compliance signals a more substantive commitment by financial institutions to decarbonization. Moreover, the framework of TCFD recommendations and the guidance issued by the FSC provide a structured pathway for reducing carbon emissions, ultimately promoting SBTi participation.

Similar to the results in Table 6, the results presented in Column (1) of Table 7 suggest that the overall TCFD compliance is positively related to the likelihood of disclosing Scope 3 emissions. Moreover, in Columns (3) to (5) of Table 7, we further find that TCFD compliance scores in strategy, risk management, metrics and targets elements positively affect financial institutions' Scope 3 disclosure. Nevertheless, in Column (6) of Table 7, the coefficient of the average TCFD compliance score by external experts is not statistically significant; thus, no clear evidence indicates that a higher extent of TCFD compliance judged by external experts is positively associated with financial institutions' Scope 3 disclosure.

Regarding Scope 1 and 2 emissions intensity, the results of Heckman two-stage regression analyses in Table 8 generally support our expectation that financial institutions with better TCFD compliance exhibit better decarbonization performance as most of the coefficients of TCFD compliance variables are positive and significant. To synthesize the results in Tables 6–8, our results suggest that better TCFD compliance further translates into substantive decarbonization practices in terms of SBTi participation, Scope 3 disclosure, and reduced Scope 1 and 2 emissions intensity. Therefore, Hypothesis 2 is generally supported.

TABLE 2 | Descriptive statistics.

Variable	Mean	Med	SD	Min	Max
<i>AvgScore</i>	45.1451	45.5250	17.3596	8.0000	80.8000
<i>AvgScore_Gov</i>	54.0799	55.5600	18.5247	12.5000	88.5200
<i>AvgScore_Str</i>	48.9853	50.4150	18.7879	4.1700	85.5600
<i>AvgScore_RM</i>	46.1358	46.6700	18.7344	3.7000	80.9500
<i>AvgScore_MT</i>	37.2095	35.2800	19.6110	2.7800	80.1300
<i>ExpAvgScore</i>	80.4403	81.3350	6.1385	63.3300	91.3500
<i>SBTi</i>	0.2625	0.0000	0.4414	0.0000	1.0000
<i>Scope 3_Disclose</i>	0.3813	0.0000	0.4872	0.0000	1.0000
<i>Scope12_Intensity</i>	0.1639	0.1191	0.1852	0.0101	1.3073
<i>Life_Ins</i>	0.2625	0.0000	0.4414	0.0000	1.0000
<i>NonLife_Ins</i>	0.2625	0.0000	0.4414	0.0000	1.0000
<i>Fin_Holding</i>	0.3750	0.0000	0.4856	0.0000	1.0000
<i>Indep_Report</i>	0.8813	1.0000	0.3245	0.0000	1.0000
<i>TCFD_Report</i>	0.7188	1.0000	0.4510	0.0000	1.0000
<i>FSize</i>	12.5358	12.8621	2.3801	5.3881	16.0232
<i>CapHold</i>	0.1417	0.0730	0.1267	0.0212	0.6216
<i>Profitability</i>	0.0154	0.0072	0.0713	−0.6676	0.4620

Note: For Scope 1 and 2 intensity variable (*Scope12_Intensity*), the descriptive statistics are calculated based on 107 observations with Scope 1 and 2 emissions disclosures. The remaining variables are calculated for the entire sample of 160 firms.

TABLE 3 | Correlations among TCFD compliance level and decarbonization practices variables.

Variable	<i>AvgScore</i>	<i>AvgScore_Gov</i>	<i>AvgScore_Str</i>	<i>AvgScore_RM</i>	<i>AvgScore_MT</i>	<i>ExpAvgScore</i>
<i>SBTi</i>	0.5661***	0.4791***	0.5274***	0.5272***	0.5279***	0.5878***
<i>Scope 3_Disclose</i>	0.5268***	0.3757***	0.5061***	0.4996***	0.5471***	0.4104***
<i>Scope12_Intensity</i>	−0.2877***	−0.3258***	−0.2964***	−0.2398**	−0.1952**	−0.3477***

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The calculations of the correlations between Scope 1 and 2 intensity (*Scope12_Intensity*) and TCFD compliance level variables are calculated based on the subsample of 107 observations with Scope 1 and 2 emissions disclosures. The abbreviations and definitions of variables are summarized in Table 1.

4.2.3 | Extended Analyses

We present the results regarding the effects of lagged SBTi participation, Scope 3 disclosure, and other explanatory variables on TCFD compliance in Table 9. In Column (1) of Table 9, both SBTi participation and Scope 3 disclosure indicators still exert positive effects on the overall TCFD compliance score. Nevertheless, in Columns (2) to (6), many of the coefficients of these two indicators lose statistical significance. In Column (6) of Table 10, we find that the effect of the lagged value of the average TCFD compliance score by external experts still exerts a positive and significant effect on SBTi participation. Similarly, in Column (1) of Table 11, we find that the coefficient of the lagged overall TCFD compliance score is positive and marginally significant in the analyses of Scope 3 emissions disclosure. Moreover, in Columns (3) to (5) of Table 11, the coefficients of the lagged values of TCFD compliance scores in strategy, risk management, metrics and targets elements are also positive and

marginally significant. Collectively, after lagging the variables of interest and other explanatory variables in the regression analyses, Hypotheses 1 and 2 are partially supported.

We present the results of the regression analyses on the matched samples constructed by the PSM method in Table 12. In Column (1) of Table 12, the treated and control groups refer to SBTi participants and non-participants, respectively. In Column (2) of Table 12, the treated and control groups refer to the financial institutions with and without Scope 3 emissions disclosure, respectively. In Columns (3) and (4) of Table 12, the treated and control groups refer to the financial institutions whose overall TCFD compliance score is above and below the median value, respectively. The details of the balance test results can be found in Appendix A.

In Panel A of Table 12, we find that both SBTi participation and Scope 3 disclosure indicators exert positive effects on the

TABLE 4 | Effects of SBTi and Scope 3 emissions disclosure on TCFD compliance level—OLS analyses on whole sample.

Variable	Column (1): AvgScore		Column (2): AvgScore_Gov		Column (3): AvgScore_Str		Column (4): AvgScore_RM		Column (5): AvgScore_MT		Column (6): ExpAvgScore	
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	-15.5036	11.7037	-26.0622*	14.9940	3.0007	13.9771	-4.5900	13.5360	-32.0521**	14.0158	66.6067***	4.6482
SBTi	6.4493**	2.6843	7.3995**	3.4390	5.6543*	3.2057	6.8365**	3.1046	4.7937	3.2146	2.8875**	1.0661
Scope3_dis	5.7119***	2.1232	1.2785	2.7200	5.9873**	2.5356	5.1892**	2.4555	9.4518***	2.5426	1.3820	0.8432
Fin_Holding	3.0584	2.4346	1.2797	3.1190	6.0468**	2.9074	3.2594	2.8157	1.7713	2.9155	2.8489***	0.9669
Indep_Report	1.3333	2.8653	9.7300***	3.6709	1.1586	3.4219	2.0130	3.3139	-3.4257	3.4314	-1.4551	1.1380
TCFD_Report	5.8769***	2.0298	12.3277***	2.6004	3.4710	2.4240	6.2263***	2.3475	4.9805**	2.4307	0.8561	0.8061
Life_Ins	-7.7616***	2.1807	-0.6761	2.7938	-8.7262***	2.6043	-11.0980***	2.5221	-9.2738***	2.6115	-2.4413***	0.8661
NonLife_Ins	8.3123**	4.1783	9.9509*	5.3529	8.3096*	4.9899	7.0671	4.8324	9.0568*	5.0037	8.1067***	1.6594
FSize	4.2604***	0.7932	4.6983***	1.0162	3.3927***	0.9473	3.6679***	0.9174	5.1782***	0.9499	1.0562***	0.3150
CapHold	-22.1383	14.4220	-8.1417	18.4764	-40.6114**	17.2234	-37.6440**	16.6798	-11.5671	17.2711	-17.8476***	5.7278
Profitability	-11.9727	12.1956	-24.7888	15.6241	-15.1572	14.5645	-7.0201	14.1049	-2.7279	14.6049	-3.3655	4.8435
F-Test	28.74***		15.38***		20.94***		23.10***		23.94***		19.70***	
Adj. R-Square	0.6357		0.4749		0.5564		0.5816		0.5906		0.5404	

Note. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The abbreviations and definitions of variables are summarized in Table 1.

TABLE 5 | Effects of decarbonization practices variables on TCFD compliance level—OLS analyses on the subsample of firms with Scope 1 and 2 emissions disclosures.

Variable	Column (1): AvgScore		Column (2): AvgScore_Gov		Column (3): AvgScore_Str		Column (4): AvgScore_RM		Column (5): AvgScore_MT		Column (6): ExpAvgScore	
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	-8.9962	18.9593	-10.5892	22.5737	26.6773	22.3609	0.1770	22.3454	-37.5492	23.8053	66.3669	6.5376
Scope12_Intensity	-21.9474***	6.6803	-25.6899***	7.9538	-26.9641***	7.8789	-24.4352	7.8734	-11.2050	8.3878	-8.1246	2.3035
SBTi	5.5044*	3.0977	9.4891**	3.6882	3.5736	3.6534	5.1936	3.6509	2.6735	3.8894	2.9510	1.0681
Scope3_dis	5.7809**	2.4397	2.2428	2.9049	5.3413*	2.8775	5.1106	2.8755	9.0923	3.0633	1.9619	0.8413
Fin_Holding	1.0314	2.9529	-0.8391	3.5159	4.8931	3.4827	1.3605	3.4803	0.2402	3.7077	0.6629	1.0182
Indep_Report	4.8372	4.3742	15.9797***	5.2081	-1.3582	5.1590	3.1110	5.1554	1.4830	5.4922	1.0475	1.5083
TCFD_Report	6.7951***	2.4009	12.2720***	2.8586	5.5889*	2.8316	7.4486	2.8297	5.4986	3.0146	1.9307	0.8279
Life_Ins	-12.2345***	2.7456	-3.9401	3.2691	-14.0165***	3.2383	-16.4645	3.2360	-13.1039	3.4474	-3.8992	0.9468
NonLife_Ins	2.3149	7.1231	-1.1240	8.4810	1.2496	8.4011	-2.7789	8.3953	11.6271	8.9437	8.2500	2.4562
FSize	3.9063***	1.2640	3.3437**	1.5049	2.3176	1.4908	3.5837	1.4897	5.6770	1.5870	1.0154	0.4358
CapHold	-0.0351	20.3271	25.8406	24.2023	-19.9995**	23.9741	-1.6424	23.9575	-17.0235	25.5227	-10.5007	7.0092
Profitability	-92.7506	62.6098	-86.6943	74.5457	-137.2514*	73.8430	-73.3866	73.7918	-86.5890	78.6129	-60.2873	21.5892
F-Test	17.84***		12.05***		13.04***		14.31***		13.27***		16.69***	
Adj. R-Square	0.6361		0.5342		0.5554		0.5801		0.5602		0.6590	

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The abbreviations and definitions of variables are summarized in Table 1. The analyses involve 107 observations with Scope 1 and 2 emissions disclosures.

TABLE 6 | Effect of TCFD compliance level on SBTi participation—Logit regression analyses on whole sample.

Variable	Column (1)		Column (2)		Column (3)		Column (4)		Column (5)		Column (6)	
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	−22.5962***	5.7049	−23.4195***	5.7084	−23.1689***	5.6359	−22.9640***	5.7242	−18.9662***	5.0321	−51.1394***	11.2024
AvgScore	0.1129***	0.0356										
AvgScore_Gov			0.0639***	0.0230								
AvgScore_Str					0.0892***	0.0316						
AvgScore_RM							0.1026***	0.0336				
AvgScore_MT									0.0569**	0.0243		
ExpAvgScore											0.4306***	0.1167
Fin_Holding	2.5257***	0.6973	2.5327***	0.6572	2.2695***	0.6533	2.5006***	0.6775	2.6056***	0.6524	1.8570***	0.7127
Life_Ins	1.5127*	0.9139	0.3888	0.7546	1.0579	0.8277	1.6748*	0.9531	0.7214	0.8096	1.0269	0.8413
NonLife_Ins	4.4955**	2.2494	4.3409**	2.2760	5.0277**	2.2657	5.6356**	2.3809	4.1302**	2.1891	1.6657	2.6216
FSize	0.9531***	0.3526	1.2046***	0.3480	1.0870***	0.3395	1.0213**	0.3482	0.9705**	0.3402	0.9010***	0.3352
CapHold	0.9621	9.4380	1.1444	8.6927	−0.7117	9.3157	−2.7013	9.4841	−0.1909	8.7353	4.2470	10.8048
Profitability	−10.7207**	4.8435	−10.7483**	4.9269	−10.1046**	4.8368	−10.2887**	4.9301	−11.2653**	4.5186	−9.0956**	4.2383
Chi-Square Test	109.88***		105.90***		106.10***		108.60***		102.48***		116.37***	
Pseudo R-Square	0.5965		0.5749		0.5760		0.5895		0.5563		0.6317	

Note. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The abbreviations and definitions of variables are summarized in Table 1.

TABLE 7 | Effect of TCFD compliance level on scope 3 emissions disclosure—Logit regression analyses on whole sample.

Variable	Column (1)		Column (2)		Column (3)		Column (4)		Column (5)		Column (6)	
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	−5.5805***	2.7208	−6.3107**	2.6708	−6.2790**	2.7112	−5.9929***	2.6979	−4.3754	2.7603	−11.2820***	4.1830
AvgScore	0.0512***	0.0195										
AvgScore_Gov			0.0161	0.0131								
AvgScore_Str					0.0437**	0.0177						
AvgScore_RM							0.0377**	0.0169				
AvgScore_MT									0.0460***	0.0154		
ExpAvgScore											0.0855	0.0526
Fin_Holding	0.3064	0.4836	0.5838	0.4647	0.2968	0.4863	0.4088	0.4774	0.4126	0.4731	0.3213	0.5080
Life_Ins	−0.7292	0.5069	−1.2091***	0.4661	−0.7799	0.5050	−0.7586	0.5174	−0.7052	0.5025	−1.0428**	0.4860
NonLife_Ins	1.6973	1.2637	1.8643	1.2121	1.7509	1.2569	1.8528	1.2318	1.6135	1.3140	1.3294	1.2550
FSize	0.2606	0.2091	0.4410**	0.1957	0.3232	0.2006	0.3345*	0.2020	0.2163	0.2113	0.3664*	0.1995
CapHold	−8.4852	5.1803	−8.9700*	4.8236	−8.0186	5.2171	−8.4399*	5.0465	−8.8875	5.4182	−8.4925*	4.9811
Profitability	2.0429	6.7633	1.4451	7.3256	2.1916	6.5456	1.1517	6.3478	1.3599	7.2970	1.4909	6.5579
Chi-Square Test	57.89***		51.80***		56.97***		55.62***		59.92***		53.05***	
Pseudo R-Square	0.2722		0.2436		0.2678		0.2615		0.2817		0.2494	

Note. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The abbreviations and definitions of variables are summarized in Table 1.

TABLE 8 | Effect of TCFD compliance level on the amount of Scope 1 and 2 emissions, Heckman two-stage regression analyses.

Variable	Stage 1: Dependent variable: <i>Scope12_dis</i>				Stage 2: Dependent variable: <i>Scope12_Intensity</i>											
	Column (1)		Column (2)		Column (3)		Column (4)		Column (5)		Column (6)					
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	-5.0399***	1.6669	1.0255***	0.2626	1.1624***	0.2666	1.0444***	0.2610	1.0362***	0.2635	1.0006***	0.2770	1.8186***	0.3553		
<i>AvgScore</i>			-0.0037***	0.0013												
<i>AvgScore_Gov</i>					-0.0028***	0.0010										
<i>AvgScore_Str</i>							-0.0036***	0.0011								
<i>AvgScore_RM</i>									-0.0031***	0.0011						
<i>AvgScore_MT</i>																
<i>ExpAvgScore</i>											-0.0011	0.0011			-0.0119***	0.0037
<i>SBTi</i>			0.0491	0.0412	0.0521	0.0420	0.0452	0.0401	0.0434	0.0407	0.0208	0.0421	0.0657	0.0421		
<i>Fin_Holding</i>	0.0430	0.3128														
<i>Indep_Report</i>	1.5262***	0.3694														
<i>TCFD_Report</i>	-0.1608	0.2746														
<i>Life_Ins</i>	-0.5085*	0.2938	-0.2215***	0.0399	-0.1860***	0.0383	-0.2238***	0.0396	-0.2296***	0.0411	-0.2029***	0.0424	-0.2182***	0.0390		
<i>NonLife_Ins</i>	0.8075	0.5675	-0.2528***	0.0934	-0.2902***	0.0933	-0.2352***	0.0933	-0.2689***	0.0934	-0.2603***	0.0983	-0.1546	0.0999		
<i>FSize</i>	0.3517***	0.1100	-0.0455**	0.0191	-0.0582**	0.0183	-0.0458**	0.0188	-0.0479**	0.0190	-0.0546***	0.0204	-0.0463**	0.0187		
<i>CapHold</i>	-1.0692	1.9528	-0.0123 s	0.2836	0.0787	0.2833	-0.0876	0.2843	-0.0121	0.2846	0.0076	0.2973	-0.1380	0.2859		
<i>Profitability</i>	-0.0119	2.0878	-0.7639	0.8614	-0.6054	0.8633	-0.9757	0.8575	-0.6652	0.8595	-0.6095	0.8880	-1.1609	0.8723		
<i>Inverse mills ratio</i>			0.0418	0.0706	0.0210	0.0725	0.0734	0.0585	0.0499	0.0704	0.0648	0.0726	0.0375	0.0700		
<i>Wald-Test</i>	—		67.11***		66.03***		70.23***		66.33***		54.91***		70.44***			

Note. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. In the first stage of Heckman two-stage regression analyses, the dependent variable is *Scope 1 and 2 disclosure indicator (Scope12_dis)*, which equals 1 if firm discloses *Scope 1 and 2 emissions*, and 0 otherwise. In the second stage, the dependent variable is *Scope 1 and 2 emission intensity (Scope12_Intensity)*.

TABLE 9 | Effects of decarbonization practices variables on TCFD compliance level—Lagged values of explanatory variables.

Variable	Column (1): AvgScore		Column (2): AvgScore_Gov		Column (3): AvgScore_Str		Column (4): AvgScore_RM		Column (5): AvgScore_MT		Column (6): ExpAvgScore	
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	−20.4260	15.0629	−20.9848	20.2439	−4.0876	18.5900	−15.6067***	16.6546	−30.3189*	16.6106	63.7701***	5.7381
SBTi	8.1262**	3.6782	11.0980**	4.9433	6.7631	4.5394	7.2957*	4.0668	6.1639	4.0561	0.8357	1.4012
Scope3_dis	6.3805**	2.7438	4.6540	3.6875	4.7146	3.3863	4.0177	3.0337	11.6311***	3.0257	1.0106	1.0452
Fin_Holding	3.0881	3.1639	−1.8957	4.2521	7.7867**	3.9047	4.6620	3.4982	2.2888	3.4889	3.4214***	1.2052
Indep_Report	−0.7937	3.7762	4.2301	5.0751	1.1519	4.6605	1.9098	4.1753	−6.9471	4.1642	−1.1050	1.4385
TCFD_Report	7.5103***	2.7493	15.4512***	3.6949	6.3814*	3.3931	8.4814***	3.0398	3.1301	3.0318	1.9768*	1.0473
Life_Ins	−7.2362**	2.8874	3.7226	3.8805	−8.8319**	3.5635	−10.0819***	3.1925	−12.6422***	3.1840	−2.3871**	1.0999
NonLife_Ins	5.5192	5.2410	5.1143	7.0437	3.0061	6.4682	3.0833	5.7948	11.0728*	5.7795	5.0699**	1.9965
FSize	4.7755***	1.0231	4.5411***	1.3750	3.8141***	1.2627	4.4530***	1.1312	5.5725***	1.1283	1.3489***	0.3898
CapHold	−16.7532	17.6057	−0.8543	23.6612	−27.5685	21.7282	−26.5676	19.4661	−25.6497	19.4147	−14.4386**	6.7067
Profitability	29.9106	21.2665	34.3358	28.5812	32.5613	26.2462	34.4494	23.5137	14.2673	23.4516	20.6580**	8.1012
F-Test	18.82***		8.79***		12.26***		16.30***		20.70***		11.66***	
Adj. R-Square	0.6929		0.4965		0.5876		0.6595		0.7138		0.5744	

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All the explanatory variables are lagged for 1 year; thus, the sample size is reduced to 80 observations. The abbreviations and definitions of variables are summarized in Table 1.

TABLE 10 | Effects of TCFD compliance level variable on SBTi participation—The use of lagged values of explanatory variables.

Variable	Column (1)		Column (2)		Column (3)		Column (4)		Column (5)		Column (6)	
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	−18.2358***	6.6454	−20.2104***	7.2354	−17.7725***	6.4602	−18.2388***	6.6075	−17.2129***	6.3379	−59.0872***	19.7844
<i>AvgScore</i>	0.0363	0.0403										
<i>AvgScore_Gov</i>			0.0360	0.0293								
<i>AvgScore_Str</i>					0.0213	0.0358						
<i>AvgScore_RM</i>							0.0317	0.0371				
<i>AvgScore_MT</i>									0.0039	0.0304		
<i>ExpAvgScore</i>											0.5649***	0.2096
<i>Fin_Holding</i>	2.7049***	0.8672	2.6504***	0.8610	2.7866***	0.8684	2.7388***	0.8609	2.8982***	0.8610	1.3467	1.1297
<i>Life_Ins</i>	0.4549	1.1785	0.4120	1.0748	0.1678	1.0960	0.5315	1.2480	−0.1001	1.0349	2.5168	1.6728
<i>NonLife_Ins</i>	2.1521	2.8701	2.6705	3.0176	2.3843	2.9388	2.3524	2.7978	2.7639	3.0642	−4.2889	3.8255
<i>FSize</i>	1.0358**	0.4315	1.1716***	0.4418	1.0612**	0.4285	1.0494**	0.4258	1.1086**	0.4469	0.7520	0.5186
<i>CapHold</i>	6.7333	13.2351	6.2613	12.9080	5.3497	13.0238	6.1564	13.1041	3.0862	12.2181	25.5353	17.6240
<i>Profitability</i>	−50.7482	41.9573	−53.6530	47.7390	−50.4636	42.2979	−51.4322	42.3405	−48.7643	41.5499	−83.7992	65.4945
<i>Chi-Square Test</i>	59.79***		60.51***		59.30***		59.71***		58.95***		71.49***	
<i>Pseudo R-Square</i>	0.5926		0.5998		0.5877		0.5918		0.5843		0.7046	

Note: *** and ** indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All the explanatory variables are lagged for 1 year; thus, the sample size is reduced to 80 observations. The abbreviations and definitions of variables are summarized in Table 1.

TABLE 11 | Effects of TCFD compliance level variable on Scope 3 disclosure—The use of lagged values of explanatory variables.

Variable	Column (1)		Column (2)		Column (3)		Column (4)		Column (5)		Column (6)	
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	−8.7492**	4.4087	−8.8963**	4.2880	−8.9250**	4.4251	−8.8426**	4.3607	−7.4962*	4.3726	−13.7653***	6.8601
<i>AvgScore</i>	0.0549*	0.0307										
<i>AvgScore_Gov</i>			0.0125	0.0217								
<i>AvgScore_Str</i>					0.0497*	0.0268						
<i>AvgScore_RM</i>							0.0445*	0.0255				
<i>AvgScore_MT</i>									0.0403*	0.0235		
<i>ExpAvgScore</i>											0.0975	0.0951
<i>Fin_Holding</i>	0.0366	0.7983	0.3947	0.7660	0.0955	0.7891	0.1516	0.7797	0.2340	0.7732	0.0532	0.8728
<i>Life_Ins</i>	−1.0031	0.7884	−1.5444**	0.7292	−1.0078	0.7885	−0.9489	0.8076	−1.2496*	0.7461	−1.3079*	0.7818
<i>NonLife_Ins</i>	4.2912	2.7191	4.9487	2.6465	3.8145	2.6222	4.5055*	2.7101	4.1813	2.6427	3.5898	2.7954
<i>FSize</i>	0.5534*	0.3244	0.7233**	0.3044	0.5553*	0.3215	0.5910*	0.3159	0.5360	0.3295	0.5613*	0.3370
<i>CapHold</i>	−6.5699	8.0708	−9.3901	7.4933	−5.1607	8.1612	−7.3520	7.9418	−7.8150	7.8739	−7.6696	7.7375
<i>Profitability</i>	−67.9988*	40.5759	−65.8261*	39.1875	−63.6073	39.1700	−68.0366	41.7414	−61.1601*	36.8448	−64.1437*	38.8206
<i>Chi-Square Test</i>	45.68***		42.46***		45.89***		45.26***		45.29***		43.22***	
<i>Pseudo R-Square</i>	0.4148		0.3856		0.4168		0.4129		0.4113		0.3926	

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All the explanatory variables are lagged for 1 year, and thus the sample size is reduced to 80 observations. The abbreviations and definitions of variables are summarized in Table 1.

TABLE 12 | Results of matched sample analyses.

Variable	Panel A: The effects of decarbonization practices on TCFD compliance				Panel B: The effects of TCFD compliance on decarbonization practices			
	Column (1): Dependent variable: <i>AvgScore</i>		Column (2): Dependent variable: <i>AvgScore</i>		Column (3): Dependent variable: <i>SBTi</i>		Column (4): Dependent variable: <i>Scope3_dis</i>	
	Coeff.	SD	Coeff.	SD	Coeff.	SD	Coeff.	SD
Constant	29.9554	28.0878	−43.0905*	22.2700	−36.0134	4814.0370	−13.0532***	5.2908
<i>SBTi</i>	5.9310**	2.6788	—	—	—	—	—	—
<i>Scope3_dis</i>	—	—	6.7767***	2.4292	—	—	—	—
<i>AboveScore</i>	—	—	—	—	1.8906*	1.1450	1.6244**	0.6931
<i>Fin_Holding</i>	−8.8984**	3.5946	2.1035	3.6809	3.7809**	1.5219	1.7824*	0.9446
<i>Indep_Report</i>	7.1476	4.2601	−0.4003	4.4090	—	—	—	—
<i>TCFD_Report</i>	14.6754***	4.3644	10.6058***	2.8703	—	—	—	—
<i>Life_Ins</i>	−14.5683***	3.7384	−11.0249***	3.4189	14.0251	4814.0290	−2.4695**	0.9838
<i>NonLife_Ins</i>	−2.0914	9.3256	−1.4299	8.0121	18.9621	4814.0290	3.8411**	1.9386
<i>FSize</i>	1.4116	1.8022	5.9473***	1.4795	1.2860**	0.5649	0.9909**	0.3928
<i>CapHold</i>	−29.1152	38.8067	59.6239*	34.7778	−4.1633	11.9963	−15.7948*	8.3619
<i>Profitability</i>	24.8050	19.4892	−88.5423**	43.3464	−13.3262**	7.8887	17.8732	12.0878
<i>Sample size</i>	44		70		72		72	
<i>F-Test</i>	11.34***		9.94***		—		—	
<i>Adj. R-Square</i>	0.6840		0.5383		—		—	
<i>Chi-Square Test</i>	—		—		34.55***		18.97***	
<i>Pseudo R-Square</i>	—		—		0.5324		0.2070	

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The details of the balance tests regarding the construction of the matched samples are available in Tables A1 and A2. The abbreviations and definitions of variables are summarized in Table 1.

overall TCFD compliance score. From another perspective, we also document positive effects of overall TCFD compliance on both SBTi participation and Scope 3 disclosure. After addressing possible endogeneity concerns by forming the matched samples for regression analyses by employing the PSM method, both Hypotheses 1 and 2 are supported.

5 | Conclusion

Extending the literature on sustainability reporting and climate-related financial disclosures, we document a mutual, positive relationship between TCFD compliance and decarbonization practices in our empirical results, and such bi-directional relationship yields the following theoretical implications. First, consistent with the perspective of signaling theory, financial institutions with stronger decarbonization commitments utilize climate-related financial disclosures as strategic tools to communicate with investors and other stakeholders, thereby differentiating themselves from competitors. Expounding on this point, we observe that financial institutions with superior TCFD

compliance are associated with better decarbonization practices in terms of SBTi participation, Scope 3 emission disclosure, and reduced Scope 1 and 2 emissions intensity. Second, in combination with the mandatory disclosure regime, legitimacy, and institutional theories, the enhanced climate-related financial disclosures are likely to facilitate meaningful decarbonization actions due to increased pressures arising from public attention and stakeholders' scrutiny on firms.

Our results further yield the following policy implications. First, consistent with Fiechter et al. (2022) and Aboud et al. (2024), mandating climate-related financial disclosure seems to be a key to increasing public awareness and stakeholder scrutiny on financial institutions' climate-related financial disclosure and decarbonization practices, thus reducing the decoupling of disclosure and decarbonization commitments. Second, to further the notion of increased public awareness and stakeholder scrutiny, apart from mandating climate-related financial disclosure, to strengthen the positive bi-directional relationship between climate-related financial disclosure compliance and decarbonization commitments, regulators may further increase

institutional pressures, public awareness, and stakeholder scrutiny on firms by taking additional measures. These measures include providing a public disclosure platform, standardizing disclosure requirements to increase the comparability of reports, mandating a third-party audit or assurance on the reports, and promoting and rewarding best practices among industry peers.

Although Taiwan has mandated financial institutions' climate-related financial disclosure, such as the problems mentioned by Morrison et al. (2024), clear regulatory guidance and requirements on the contents of reports remain scarce. Thus, the disclosure and non-disclosure of certain content by financial institutions are arbitrary, with no standardized or harmonized guidance about target selection. The above-mentioned measures can address the problems associated with the absence of clear and standardized guidance, and further shape the institutional environments that foster disclosure quality and genuine commitments.

The study's limitations and recommendations for future research are outlined below. First, the sample employed in our empirical analysis is constrained to a single-country and single-industry context. Given that regulatory pressures and transitional risks vary significantly across industries and countries, future research could benefit from conducting cross-industry and cross-country analyses to provide broader insights. Second, some financial institutions in our sample are not publicly listed, resulting in limited availability of data on corporate governance, market competition, and the variables representing some important corporate decisions. The relevant recommendations are explained as follows.

Regarding the role of corporate governance, previous studies have underscored that corporate governance is a critical factor that may influence both climate-related financial disclosures and environmental performance (Luo and Tang 2021; Cosma et al. 2022). Future studies can incorporate the following factors relevant to corporate governance. The first one is institutional investors' ownership. Specifically, Aibar-Guzmán et al. (2024) suggest that long-term institutional investors' ownership positively affects the effectiveness of climate governance, and improved climate governance further leads to better financial performance. The second one is board composition. Regarding the relevant evidence, Tahat and Hassanein (2024) suggest that female board members and sustainability committees are crucial in promoting ESG performance. Hassanein and Elsayed (2025) further document the moderating role of female directors in relation to the effect of ESG reporting on stock liquidity. Moreover, Raimo et al. (2025) analyze the effects of board characteristics on the level of climate change disclosures, and they find that board expertise, gender diversity, and board size play roles in facilitating climate change disclosures. Third, regarding the evidence on cross-country samples, Elmaghrabi et al. (2025) underscore the importance of country and firm levels of sustainability governance in corporate sustainability performance. Fourth, García-Sánchez et al. (2024) suggest that climate governance is vital in facilitating innovation against climate change.

Regarding market competitions and corporate decisions, Hassanein and Elmaghrabi (2025) and Abdelbaky et al. (2025) suggest that the pressures from market competitions and firms'

earnings management practices may hinder ESG disclosure and practices. Taken together, exploring whether the TCFD compliance–decarbonization relationship is mediated and moderated by firms' corporate governance practices, market competitions, and other corporate decisions represents a promising avenue for future research.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Endnotes

¹ The full list of all 38 domestic banks is available at the website of Banking Bureau of FSC: https://www.banking.gov.tw/en/home.jsp?id=24&parentpath=0,100&mcustomize=FscSearch_BankType.jsp&type=3 (August 31, 2025).

² In the assessment conducted by the Center for Business Sustainability, in addition to the 21 non-life insurers in our analysis, two alien reinsurers are classified as non-life insurers. However, as financial data on these two alien reinsurers is not available, they were excluded from our sample. The full lists of life insurers, non-life insurers, and reinsurers are available at the website of Insurance Bureau of FSC: <https://www.ib.gov.tw/en/home.jsp?id=102&parentpath=0,7> (August 31, 2025).

³ In insurance operations, insurers typically hold a substantial amount of invested assets. Therefore, the supplemental guidance for asset owners is also applicable to insurers when assessing their compliance with the TCFD recommendations.

⁴ The main purpose of including the bonus question set is to encourage financial institutions to disclose more detailed and financially relevant quantitative information beyond the basic requirements specified in TCFD recommendations. As the disclosures in the governance core element mainly include qualitative descriptions about how climate-related risks and opportunities are governed and monitored without much quantitative information, the coding taskforce decided not to incorporate the governance core element in the bonus question set.

⁵ According to the Climate-related Financial Disclosures Yearbook, 2023 published by Center for Business Sustainability at NCCU, the main reason for some financial institutions not to disclose Scope 1 and 2 emissions is the technical difficulties in collecting data. Moreover, several financial institutions affiliated to financial holding companies disclose Scope 3 emission at financial holding company level rather than entity level. These financial institutions' Scope 1 and 2 emissions are not included in our analyses.

⁶ The selection of these control variables is primarily based on data availability consideration. In our sample, some financial institutions are not publicly listed. Thus, some market-based variables and the variables representing corporate governance practices are not available for these non-listed firms.

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Appendix A

Balance Tests Results

Tables A1 and A2 present the balance test results from the propensity score matching used to construct the matched samples.

TABLE A1 | Balance tests regarding decarbonization practices participation status.

Variable	Panel A: SBTi participants versus non-participants			Panel B: Firms with and without Scope 3 emission disclosure		
	Treated (N=22)	Control (N=22)	t-test	Treated (N=35)	Control (N=35)	t-test
<i>Fin_Holding</i>	0.7727	0.7727	0.00	0.3143	0.4857	−1.47
<i>Indep_Report</i>	0.8636	0.8182	0.40	0.9143	0.8000	1.36
<i>TCFD_Report</i>	0.9091	0.8182	0.87	0.6857	0.6857	0.00
<i>Life_Ins</i>	0.2727	0.2727	0.00	0.1714	0.1714	0.00
<i>NonLife_Ins</i>	0.2727	0.2727	0.00	0.2000	0.2000	0.00
<i>FSize</i>	13.3550	13.0780	0.40	12.9430	13.3450	−1.04
<i>CapHold</i>	0.1139	0.1304	−0.52	0.1067	0.1070	−0.02
<i>Profitability</i>	0.0031	−0.0142	0.54	0.0134	0.0141	−0.10

Note: In Panel A, treated and control groups refer to firms with and without SBTi participation, respectively; in Panel B, treated and control groups refer to firms with and without Scope 3 emissions disclosure, respectively. The abbreviations and definitions of variables are summarized in Table 1.

TABLE A2 | Balance test: High versus low compliance.

Variable	Treated (N=36)	Control (N=36)	t-test
<i>Fin_Holding</i>	0.3611	0.3611	0.00
<i>Life_Ins</i>	0.3889	0.3889	0.00
<i>NonLife_Ins</i>	0.2500	0.2500	0.00
<i>FSize</i>	12.2870	12.3660	−0.19
<i>CapHold</i>	0.1264	0.1189	0.31
<i>Profitability</i>	0.0107	−0.0004	0.54

Note: Treated and control groups refers to firms whose average score on overall TCFD compliance level is above and below the median value, respectively. The abbreviations and definitions of variables are summarized in Table 1.